Does a rule of thumb choose better once the tactics are checked?

The app's rules of thumb after its blunder check, in quiet positions from 127,914 Lichess positions refereed by Stockfish

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Abstract

A coach puts the checks, captures and threats first and a rule of thumb second, and so do several of the app's lessons. The previous study found that no rule of thumb chooses a better move than the one a player at his level would naturally play, mostly because the rule's move left a piece to be taken or passed over a tactic. This one tests the rule as the lessons teach it: in positions with no tactic on the board by the app's reading, and among the moves the app's blunder check passes, is the move a rule points to, as his level would play it, sound as often as the move his level naturally plays among the same moves? The rule, fixed before any position was read, first asks a control to show that the check can rescue a rule's move. It does not: where the rule's own move fails the check, the next rule move that passes it is no sounder, its soundness minus the rule move's -1.0 points on the train months and 2.3 held out, both intervals across 0, so no rule receives a verdict. The descriptions are clear all the same. In a quiet position the check rarely touches the rule's move, and that move is sound less often than the natural one for all twelve rules with data, on both periods: by 1.3 points for the rule of the square up to 32.0 for the pawn break held out. Over all positions, tactics included, the gap is 1.3 to 2.4 times its quiet size. The rules still point to sounder moves than chance, ten of them on both periods. A rule of thumb tells a player where to look and explains a move; at his level it does not choose one, even after the tactics.

1 The question

The previous study asked each of the app's rules of thumb to choose a move with no tactics checked first: among the legal moves that show the rule, the one Maia-2 rates most likely at the mover's band. No rule chose better than the natural move, the one Maia-2 rates most likely of all, and its held-out examples said why. Most of the rule's unsound moves lose material by the app's own blunder check, and most of the rest pass over a check, a capture or a promotion that the natural move took.

A coach would not give the rule that way. Several of the app's lessons begin "No tactics?", and after that study its open-file and outpost lessons put the checks, captures and threats first too. So the fair test of such a lesson is the rule after the check. In positions with no tactic on the board, and among the moves the app's blunder check passes, is the move a rule points to, as players at his level would play it, sound as often as the move they would naturally play among those same moves? If it is sounder, a lesson can teach the rule as the move to look for once nothing is hanging. If it is no sounder, the rule explains a move and says where to look, but does not choose one. If it is less sound, a lesson that teaches it should tell him to compare its move with the others before playing it.

2 How it was measured

The positions. The 127,914 positions the previous study scored for its 13 rules with a verdict, read and never changed: middlegames and endgames from rated 10-minute Lichess games with no increment, the mover rated 1250 to 1649 on Lichess (his band and the one above), with July and August 2026 to train on and September 2026 held out, at most one position a game for each rule. Each keeps the rule's legal moves, the moves that study compared and the referee's evidence for them.

The check. Two readings by the app's knowledge service at commit `3437a71184`. A position is *quiet*, the lessons' "no tactics", when the mover is not in check and neither side has a mate in one or a winning capture, by the app's description of the position. A move *passes the check* when the app's safety check calls it safe: the reply cannot mate at once, no piece of the mover's is newly left to be taken for a gain (the capturing piece of an even trade is not counted), and the move deserts no defence.

The moves compared. In each quiet position, R_s is the rule's move after the check: among the rule's moves that pass it, the one Maia-2 rates most likely at the mover's band. N_s, the natural move after the check, is Maia-2's most likely move among all the moves that pass. R_{0s} is the move Maia-2 likes best among as many passing moves as R_s had to choose from, drawn at random: a rule with no information. R and N are the previous study's rule move and natural move, before the check. Maia-2 is the app's at commit `f4bd63f19b`, the one that study measured, and every R and N was reproduced before a position was scored.

The referee. A move the previous study scored keeps its soundness, and with seven men or fewer its stored tablebase probe decides. Every other move is scored by Stockfish through the app's engine service at `f4bd63f19b`, at its grading budget, against one fresh search for the best in that position: a move is *sound* when it stays within the app's window of the best. All 84,198 fresh searches found the best that study had stored. The scoring took 9.3 worker hours on two workers.

The rule, fixed before any position was read. The *lift* is R_s's soundness minus N_s's, paired by position over the quiet positions where some rule move passes the check, with 95% intervals from 2,000 bootstrap draws over games (seed 20261018). A control comes first: pooled over the rules, in the quiet positions where R fails the check and some rule move passes it, R_s's soundness minus R's must have its interval above 0 on both periods, or no rule gets a verdict, since a lift this design could not see would read as none. Then a rule holds after the check when its lift's interval lies above 0 on the train months and on the held-out month, misleads when it lies below 0 on both, and does no better than the natural move otherwise; with fewer than 300 held-out positions there is not enough data. The *content* after the check, R_s's soundness minus R_{0s}'s, the same measures over every position with a safe rule move, quiet or not, and the lift in splits of the positions describe and change no verdict.

3 The control fails

In the quiet positions with a safe rule move, the rule's own move fails the check only rarely: 0.9% of them on the train months and 1.1% held out, which leaves the control 292 and 175 positions (Table 1). Where it fails, the next rule move that passes the check is no sounder: R_s's soundness minus R's is −1.0 points (−6.5 to 4.5) on the train months and 2.3 (−6.0 to 10.1) held out. By the rule, then, no rule receives a verdict, and everything below describes. This study does not separate the two ways the control can fail: a quiet move the check flags may be sound after all, or the next rule move may be no better.

Period	Quiet positions with a safe rule move	Control positions	Games	R_s minus R, points
Train	30,758	292	288	−1.0 (−6.5, 4.5)
Held out	15,489	175	173	2.3 (−6.0, 10.1)

Table 1: The control, pooled over the 13 rules: the quiet positions where the rule's move R fails the app's blunder check and some rule move passes it, and the soundness of R_s, the rule's move after the check, minus R's, paired by position, with its 95% interval from 2,000 draws over games. The rule asked for the interval to lie above 0 on both periods.

4 What the measures describe

The check changes little in a quiet position. Where some rule move passes it, the rule's own move passes it in 95.5% to 100.0% of the positions, for every rule on both periods, so R_s is nearly always R; the natural move passes it in 93.9% to 100.0%, and the check moves the natural move's soundness by at most 0.9 points for every rule with data. What the check removes is whole positions: the outside passed pawn has a rule move that passes it in only 8% of its quiet positions held out, which leaves 12, too few for a verdict even had the control passed, and cutting the king off, the passed pawn and the worst piece have one in 76% to 82% of theirs. How many of a rule's positions are quiet at all varies from 12% for the outside passed pawn to 72% for the opposition.

The rule's move still trails the natural move. After the check the rule's move is sound less often than the natural move, for all twelve rules with data, on the train months and on the held-out month, every interval below 0 (Table 2): by 1.3 points for the rule of the square up to 32.0 for the pawn break held out. Where the two moves differ, the rule's is sound less often by 6.7 points for the rule of the square up to 36.4 for the pawn break. This is the previous study's finding again, in the positions where its two main reasons, a piece left to be taken and a tactic passed over, are gone by the app's reading.

The rules still carry information. Against the move Maia-2 likes best among as many safe moves drawn at random, ten rules point to sounder moves on both periods, from 3.4 points for the opposition to 36.8 for trading when ahead held out. The rook behind the passed pawn does so on the train months, by 2.1 points (0.2 to 3.8), but not held out (2.1, −0.5 to 4.6), and cutting the king off reads no differently from chance, −1.1 points (−4.9 to 2.8) held out, as in the previous study.

	Held-out positions	Quiet share	R_s sound	N_s sound	Lift, train	Lift, held out	Lift, all positions	Content, held out
Active king	2,245	59	72.5	82.3	−10.7	−9.8 (−11.5, −8.2)	−23.3	7.5 (5.7, 9.4)
Opposition	1,015	72	85.8	94.3	−6.0	−8.5 (−10.6, −6.4)	−10.6	3.4 (1.3, 5.7)
Rule of the square	1,552	60	97.2	98.5	−1.2	−1.3 (−2.0, −0.6)	−2.1	10.0 (8.3, 11.6)
Rook behind the passed pawn	1,307	35	51.0	75.7	−25.3	−24.6 (−27.5, −21.6)	−41.3	2.1 (−0.5, 4.6)
Cutting the king off	552	52	53.3	82.2	−28.6	−29.0 (−33.1, −25.0)	−37.6	−1.1 (−4.9, 2.8)
Outside passed pawn	12	12	58.3	75.0	0.0	−16.7 (−42.9, 0.0)	−4.6	8.3 (−36.4, 50.0)
Trading when ahead	1,033	27	77.6	84.4	−6.9	−6.8 (−8.8, −4.8)	−11.8	36.8 (33.3, 40.2)
Open file	1,454	38	54.1	71.1	−18.0	−17.0 (−20.0, −14.0)	−30.5	7.0 (4.3, 9.6)
Rook on the seventh	1,176	31	59.5	76.7	−16.2	−17.2 (−20.1, −14.2)	−28.1	14.9 (12.0, 18.0)
Outpost	1,259	34	53.8	73.9	−20.7	−20.2 (−23.0, −17.3)	−30.3	13.0 (10.0, 16.0)
Pawn break	1,620	44	39.5	71.5	−30.5	−32.0 (−34.8, −29.1)	−47.6	3.6 (0.9, 6.4)
Passed pawn	1,031	32	64.3	79.1	−14.4	−14.8 (−17.5, −12.2)	−29.2	12.6 (9.6, 15.6)
Worst piece	1,233	37	52.6	75.1	−20.1	−22.5 (−25.5, −19.4)	−36.4	4.7 (2.0, 7.4)

Table 2: What the measures describe, with no verdict. Held-out positions: quiet ones where some rule move passes the check. Quiet share: of all the rule's held-out positions, in percent. R_s and N_s sound: the share of those positions where the rule's move and the natural move after the check are sound, in percent. The lift is R_s's soundness minus N_s's in points, with its 95% interval from 2,000 draws over games (for every rule with data the train interval lies below 0 as well); over all positions it is the same lift over every position with a safe rule move, quiet or not. The content is R_s's soundness minus that of R_{0s}, the move Maia-2 likes best among as many safe moves drawn at random.

5 With a tactic on the board

Over every position with a safe rule move, quiet or not, the gap is larger for each of the twelve rules on both periods: 1.3 to 2.4 times its size in the quiet positions, from the opposition to the active king, and 2.1 to 47.6 points held out (Table 2). The quiet positions are among them, so in the rest, where the mover is in check or either side has a mate in one or a winning capture, the gap is larger still. A tactic on the board is where a rule's move falls furthest behind the natural one, which is the order the lessons teach: the tactics first.

6 Where the gap is smallest

The splits describe and decide nothing (Table 3). For all eight rules whose move after the check is a capture or a check in some positions and neither in others, on both periods, the forcing move trails the natural move by less. Of the four rules with at least 50 forcing moves in each period, the open file and the rook on the seventh do not clearly trail when forcing, their intervals across 0 on both periods (the open file: −4.1 points forcing and −17.9 otherwise, held out). For all eight rules that come up in both phases, the gap is larger in the middlegame than in the endgame on both periods, as with the passed pawn (−25.1 points against −9.7 held out); the active king, the opposition, the rule of the square and cutting the king off come up almost only in endgames. And for nine of the eleven rules with positions ahead, level and behind, the gap is largest with level material on both periods; the exceptions are the rule of the square and cutting the king off. As in the previous study, level material may be in part the referee's doing, an inference this study does not test: a fixed window in win% is narrower in centipawns where the game is balanced.

	R_s quiet	R_s forcing	Middle-game	Endgame	Ahead	Level	Behind
Active king	−10 (2,244)	0 (1)	–	−10 (2,245)	−9 (778)	−14 (875)	−4 (592)
Opposition	−8 (1,015)	–	–	−8 (1,015)	−4 (230)	−11 (642)	−6 (143)
Rule of the square	−1 (1,549)	0 (3)	–	−1 (1,552)	−1 (1,205)	−1 (329)	−6 (18)
Rook behind the passed pawn	−25 (1,298)	11 (9)	−33 (397)	−21 (910)	−23 (587)	−31 (454)	−18 (266)
Cutting the king off	−29 (552)	–	–	−29 (552)	−16 (176)	−34 (341)	−46 (35)
Trading when ahead	–	−7 (1,033)	−8 (806)	−4 (227)	−7 (1,033)	–	–
Open file	−18 (1,356)	−4 (98)	−19 (976)	−12 (478)	−15 (351)	−19 (794)	−14 (309)
Rook on the seventh	−18 (1,113)	2 (63)	−19 (347)	−16 (829)	−12 (420)	−22 (476)	−15 (280)
Outpost	−21 (1,142)	−10 (117)	−23 (720)	−17 (539)	−19 (413)	−22 (662)	−16 (184)
Pawn break	−32 (1,619)	0 (1)	−34 (1,239)	−25 (381)	−29 (280)	−34 (1,086)	−24 (254)
Passed pawn	−16 (870)	−9 (161)	−25 (342)	−10 (689)	−13 (476)	−17 (402)	−14 (153)
Worst piece	−23 (1,210)	−9 (23)	−25 (714)	−18 (519)	−18 (312)	−27 (662)	−15 (259)

Table 3: The lift in quiet positions in each cell of a split, held out, in points, with the cell's positions in brackets; a dash is a cell with none. Forcing: R_s is a capture or gives check. The phase is the app's divider's, and material is the mover's, both read before the move by the app at the check's commit. The splits describe and decide nothing.

7 What it means for the app

Nothing is ported, since there is no verdict. The descriptions agree with the previous study's reading and narrow it. Once the checks, captures and threats are seen, the move a rule of thumb points to is still sound less often than the move his level plays naturally, and most of all with a tactic on the board. A rule tells him where to look and explains a move; it does not choose one.

That is how the app now teaches its rules. Its worst-piece lesson begins "No tactics?", its pawn-break lesson says to check what a break leaves hanging before playing it, and since chess PR 165, after the commits measured here, its open-file and outpost lessons start from the checks, captures and threats, the open-file lesson saying that the file tells him where to look, not what to play. Game Review already uses the rules that way, as the previous study found.

A round that wants a verdict needs a control with more to see. The check changes the rule's move in 0.9% and 1.1% of the quiet positions with a safe rule move, so its rescues are few; a control built from positions where a rule's move is known to leave a piece to be taken while another rule move is safe would test the check itself.

8 Caveats

- **No verdict.** The control failed, so every number here describes; none was a test the rule set.
- **Not his games.** The positions are Lichess players' in his band and the band above; his own games are too few for most rules.
- **Quiet by board code.** No mate in one and no winning capture for either side is the app's reading of "no tactics". A deeper tactic, a fork two moves away, still counts as quiet.
- **One move for a plan.** A rule is tested as the next move his level would play for it, not as a plan over several moves, and a sound move can still be the second-best plan.
- **A strong baseline.** The natural move is Maia-2's most likely move at the mover's band, which the previous study found sound more often than the move actually played.
- **Two app commits.** The check is the app's reading at `3437a71184`, after a fix to its even-trade reading; Maia-2 and the referee are the previous study's, at `f4bd63f19b`, so every R and N is

the one it scored.

- **No examples.** The rule draws held-out examples only for rules with a verdict, so this paper shows no positions.